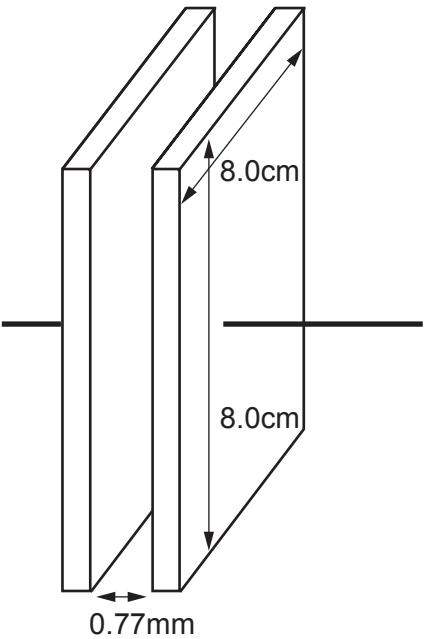


SECTION A

Answer all questions.

1. (a) (i) For the air-spaced parallel plate capacitor shown, calculate the pd applied when it stores $25.5\mu\text{J}$ of energy. [4]



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Examiner
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(ii) Explain why the capacitor stores energy when a pd is applied to the plates. [2]

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(iii) A group of scientists claim that they have developed a new dielectric that enables the above capacitor to store a million times more charge and energy for a given pd. Explain what further steps must be taken by the scientific community and industry before this new dielectric can be used in devices and sold to the public. [3]

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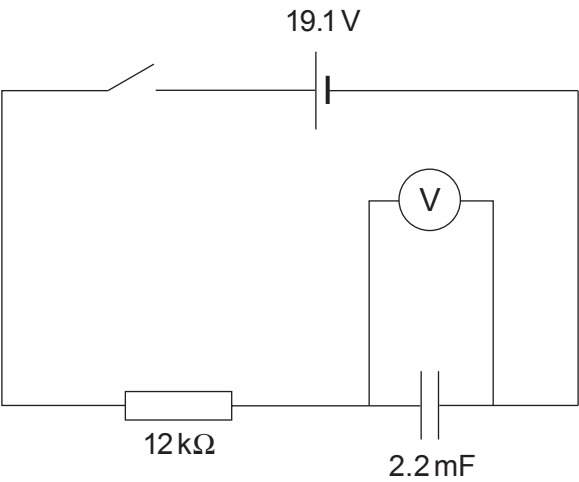
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(b) Bethan investigates the charging of a capacitor using the following circuit.



The results she obtains are then tabulated.

Time / s	pd across capacitor / V ± 0.5 V	Charge on capacitor / mC ± 1.1 mC
10.0	6.1	13.4
20.0	10.0	22.0
30.0	12.8	
40.0	15.1	33.2
50.0	16.2	35.6
60.0	17.1	
70.0	17.7	38.9
80.0	18.3	40.3

(i) Confirm that the uncertainty in the charge is 1.1 mC (you may assume that the uncertainty in the 2.2 mF capacitor is negligible). [1]

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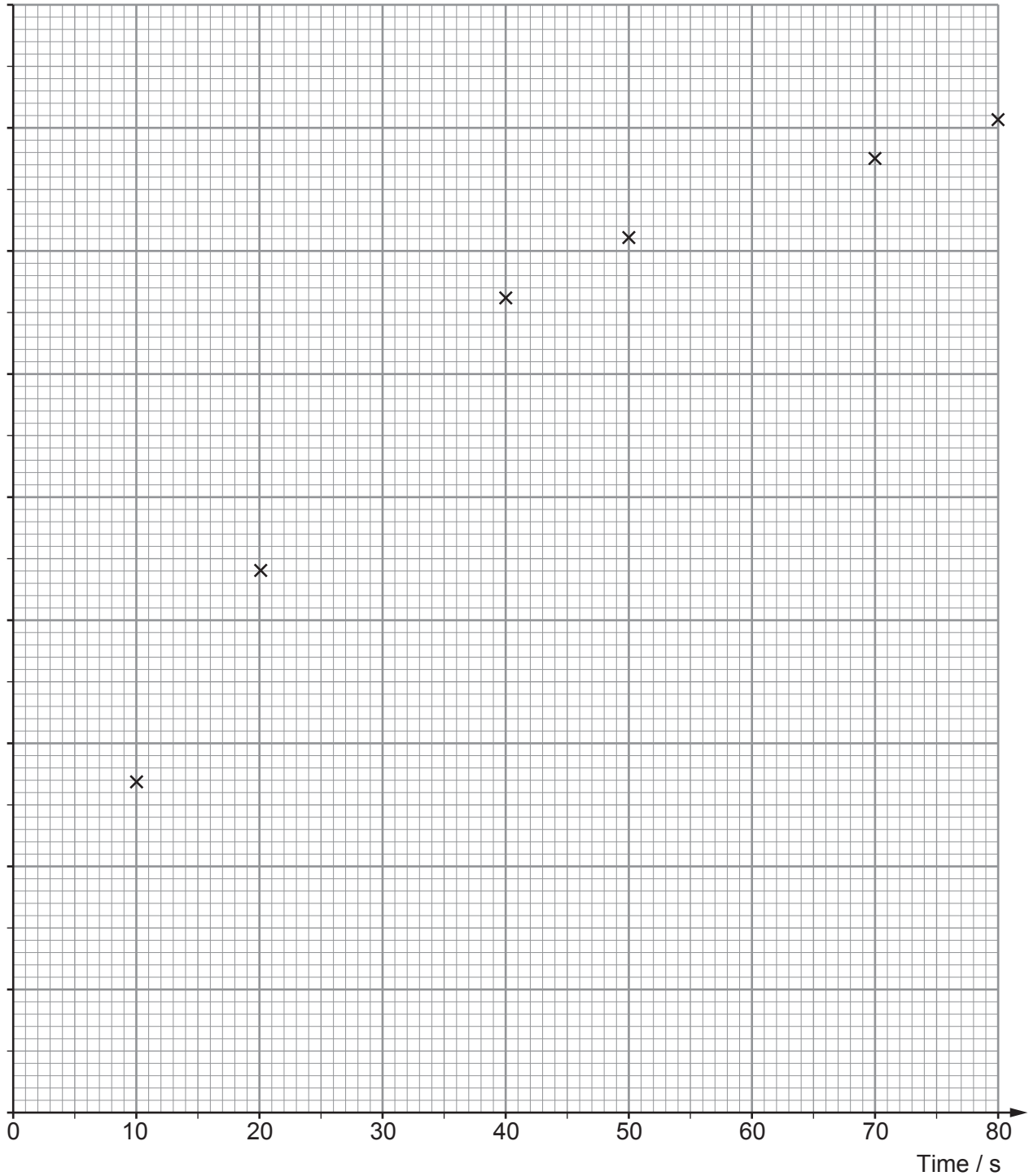
(ii) **Complete** the table. [2]



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- (iii) Complete the graph by labelling the y -axis scale, plotting the remaining two points, adding error bars for charge and drawing a curve of best fit. [5]

Charge / mC



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(iv) **Use the curve of best fit** to obtain the time constant of the charging circuit (show your working and do not simply multiply the resistance by the capacitance). [3]

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(v) By drawing a suitable tangent, calculate the current in the circuit at 45 s. [2]

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(c) Use the graph on page 5 and your answers to (b)(iv) and (b)(v) to evaluate whether or not the data obtained in this experiment are in good agreement with the equations: [5]

$$Q = Q_0 \left(1 - e^{-\frac{t}{RC}} \right) \text{ and } I = I_0 e^{-\frac{t}{RC}}$$

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